

Dhruv Bhambhani

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EDUCATION

Texas A&M University

BS in Mechatronics Engineering, Minor in AI in Business & Embedded Systems Integration

College Station, TX

Aug 2024 – May 2028

- Major GPA: 3.82 | Dean's List 2025 & 2026 | Leadership: Web Development Committee (TURTLE Robotics)

EXPERIENCE

Bank of America

Incoming Software Engineering Intern

Dallas, TX

May 2027

Powell Industries

AI Engineer, Project Lead (Full-time Remote)

Houston, TX

Aug 2026 – Present

- Spearheading an AI defeaturing pipeline to automate CAD-to-simulation transfer from Autodesk Inventor to COMSOL; scoped and greenlit the initiative directly with the VP of R&D and multiple directors.
- Authored the financial analysis and product management plan that secured project funding, defining scope, roadmap, and ROI to align R&D leadership and business-unit stakeholders.

R&D Software Engineering Intern

May 2026 – Aug 2026

- Productionized a multi-agent RAG pipeline in Snowflake using governed AI agents across all 5 R&D product lines, referencing Powell's technical docs to process 200+ queries/day with 98% accuracy, benchmarked against 10,000-query evaluation set.
- Exceeded stakeholder expectations by delivering agent-to-agent communication, automated monitoring, & self-improvement pipeline 8 weeks early, driving adoption across 50+ engineers, directors, & VPs to preserve 75 yrs of institutional knowledge.
- Prototyped local RAG pipeline using LangChain, Ollama (Llama 3), and ChromaDB with recursive character text splitting, sentence-transformer embeddings, and HNSW-indexed ANN search with cosine similarity scoring across 500+ technical docs

Secure Communication and Computer Systems (SUCCESS) Lab | Texas A&M

Lead Applied AI Researcher

College Station, TX

April 2026 – Present

- Leading novel research for multi-tiered LLM agentic architecture integrating NVIDIA Nemotron with the OpenClaw framework for multi-robot coordination across multiple ROS and ESP32 micro-ROS nodes by leveraging NLP
- Architected a security layer defending against prompt injection, context manipulation, and cross-layer exploitation in physical robotics environments based off a 470-advisory taxonomy on OpenClaw with findings contributing to ongoing research on LLMs as autonomous reasoning engines for multi-robot perception, decision-making, and physical action

Design Innovation & Generative Intelligence (DIGIT) Lab | Texas A&M

Computational AI Researcher

College Station, TX

April 2026 – Present

- Co-authoring research paper (NSF Funded, \$500,000) on applications of Interpretable Machine Learning in Mechanical Engineering by extending 1D interpretability framework to 2D interaction effects for AI-driven engineering design rules
- Directed development of a symbolic regression pipeline with PS-Tree (genetic programming/decision trees) to decode pairwise feature interactions from an Energy-Based Model predicting UTS across 100×100 effect grids
- Evaluated Ridge and linear regression as leaf-model alternatives within PS-Tree across 20 feature interactions, concluding GP-based leaves outperform for nonlinear structure-property relationships and producing publication-ready visualizations sorted by feature importance to derive design rules from neural network predictions

JBB Asset Management LLC

Software Engineering Intern

Houston, TX

May 2024 – August 2024

- Built a document-synthesis pipeline for 500+ unit portfolio using React, Python, and Flask to compile legally-compliant, customized lease agreements based on JSON-schema parameter maps, reducing contract production time by 2 days
- Engineered an asynchronous multi-agent orchestration framework leveraging Python, LangChain, and Redis queues to autonomously parse, prioritize, and dispatch maintenance tickets based on severity metrics and localized vendor constraints

PROJECTS

EDEN Robotics (Funded by Texas A&M) - Emotions/Personalities in Humanoid Robots

January 2026 – June 2026

- Built an affective AI framework enabling emotions/personalities in robots by integrating Ollama-based reasoning with Supermemory for long-term memory allowing for personalized responses based on user history and emotional context

Cryos AI - Adversarial AI Fraud Simulation | Multi-Agent Systems (2nd Place at TAMUHack26)

January 2026

- Built an adversarial multi-agent system that models historical customer behavior and known fraud techniques to simulate adaptive bank-fraud attacks and defensive countermeasures utilizing the Capital One API

Canvas MCP - Open-Source MCP Server for Canvas LMS | Python, Claude API

May 2026

- Built an open-source MCP server exposing 14 tools that give Claude Code real-time Canvas LMS data, and allows for accurate weighted grades, study materials based on course content, and on-demand flashcard generation

TECHNICAL SKILLS

Languages & Frameworks: C++, Python, Java, C, HTML/CSS, SQL, React (Next.js, Node.js), Flask, Spring Boot

AI / ML: LangChain, LlamaIndex, RAG, MCP Servers & Agents, LLM Fine-Tuning, Hugging Face Transformers, PyTorch, TensorFlow/Keras, Scikit-learn, Pandas, NumPy, Jupyter, Prompt Engineering

Data & Infra: ChromaDB, Pinecone, Weaviate, Snowflake, PostgreSQL, MongoDB, Supabase, Docker, Kubernetes, AWS, CI/CD, Git, Agile, Ollama

Certifications: Texas Real Estate License – Licensed commercial real estate professional (180 hours coursework)